

FP2 Chapter 10 Differentiation

Lecture 10 Differentiation

(Inverse Trigonometry, Hyperbolic Functions, Inverse Hyperbolic Functions, Implicit Differentiation, Parametric Differentiation, Maclaurin Series)

1. Differentiation of Inverse Trigonometry Function

- To differentiate $y = \sin^{-1} x$,

Let $\sin y = x$,

By implicit differentiation,

$$\cos y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - \sin^2 y}} = \frac{1}{\sqrt{1 - x^2}}$$

- To differentiate $y = \cos^{-1} x$,

Let $\cos y = x$,

By implicit differentiation,

$$-\sin y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = -\frac{1}{\sin y} = -\frac{1}{\sqrt{1 - \cos^2 y}} = -\frac{1}{\sqrt{1 - x^2}}$$

- To differentiate $y = \tan^{-1} x$, (shown in P3)

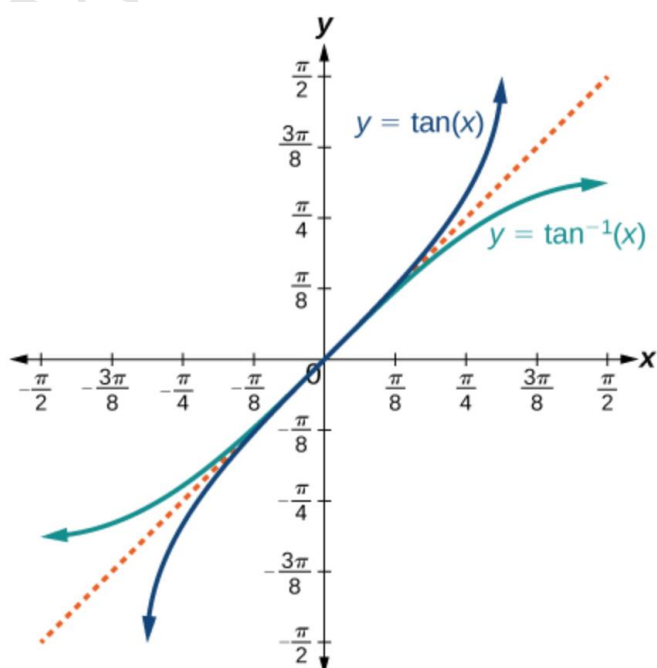
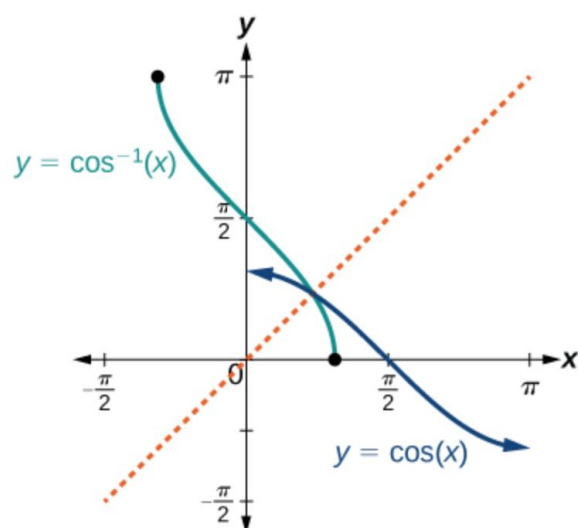
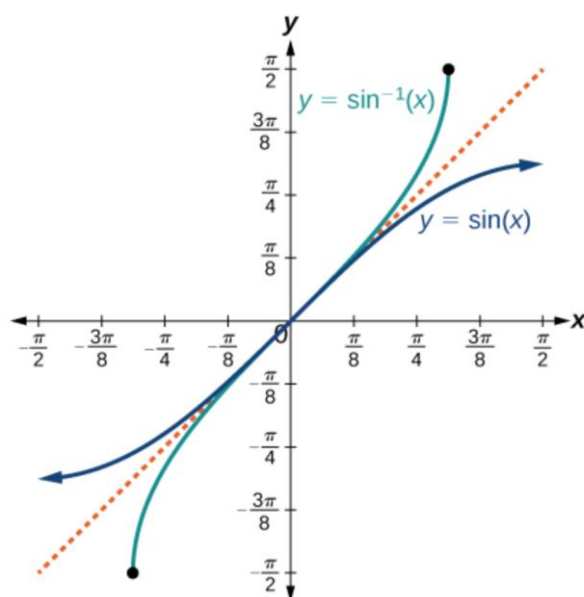
Let $\tan y = x$,

By implicit differentiation,

$$\sec^2 y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}$$

- Remember that the trigonometry functions, $\sin x$, $\cos x$ and $\tan x$ are not one-to-one function. Therefore they cannot have an inverse unless we restrict the domain.



- **Example 1: Differentiate the following functions.**

(i) $y = \sin^{-1}\left(\frac{x}{3}\right)$

(ii) $y = \cos^{-1}\sqrt{x}$

(iii) $y = (\sin^{-1} x)^2$

(iv) $y = \sin^{-1} x \cos^{-1} x$

(v) $y = \sin^{-1}(x + 3)$

(vi) $y = (\sin^{-1} 2x)^3$

- **Exercise A:**

1. Differentiate the following functions.

(i) $y = \sin^{-1}(3x)$

(ii) $y = x \sin^{-1} x$

(iii) $y = \cos^{-1}(x^2)$

(iv) $y = e^x \sin^{-1} x$

(v) $y = \sin^{-1} \sqrt{2} x$

2. The function $y = x - \sqrt{1 - x^2} \sin^{-1} x$. Show that

$$\frac{dy}{dx} = \frac{x \sin^{-1} x}{\sqrt{1 - x^2}}.$$

2. Differentiation of Hyperbolic Function

- To differentiate $y = \sinh x$,

$$y = \sinh x = \frac{1}{2}(e^x - e^{-x})$$

$$\frac{dy}{dx} = \frac{d}{dx}(\sinh x) = \frac{1}{2} \frac{d}{dx}(e^x - e^{-x}) = \frac{1}{2}(e^x - e^{-x}) = \cosh x$$

- To differentiate $y = \cosh x$,

$$y = \cosh x = \frac{1}{2}(e^x + e^{-x})$$

$$\frac{dy}{dx} = \frac{d}{dx}(\cosh x) = \frac{1}{2} \frac{d}{dx}(e^x + e^{-x}) = \frac{1}{2}(e^x - e^{-x}) = \sinh x$$

- To differentiate $y = \tanh x$,

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(\tanh x) = \frac{d}{dx} \left(\frac{\sinh x}{\cosh x} \right) = \frac{\cosh x \cosh x - \sinh x \sinh x}{\cosh^2 x} = \frac{1}{\cosh^2 x} \\ &= \operatorname{sech}^2 x \end{aligned}$$

- **Example 2: Differentiate the following functions.**

(i) $y = \operatorname{sech} x$

(ii) $y = \cosh 3x$

(iii) $y = \tanh 2x$

(iv) $y = \sinh 4x$

(v) $y = \sinh^2 3x$

(vi) $y = \sqrt{\cosh 5x}$

- **Exercise B: Differentiate the following functions.**

(i) $y = x \sinh 2x$

(ii) $y = \cosh 2x \tanh 2x$

(iii) $y = \ln(\cosh x)$

(iv) $y = e^{2x} \cosh 3x$

(v) $y = \frac{1}{2} \sinh 4x + 4 \sinh 2x + 6x$, show $\frac{dy}{dx} = 16 \cosh^4 x$.



3. Differentiation of Inverse Hyperbolic Function

- To differentiate $y = \sinh^{-1} x$,

Let $\sinh y = x$,

By implicit differentiation,

$$\cosh y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cosh y} = \frac{1}{\sqrt{1 + \sinh^2 y}} = \frac{1}{\sqrt{1 + x^2}}$$

- Alternately, you can find the derivative from the logarithmic form.

$$\begin{aligned} \frac{d}{dx}(\sinh^{-1} x) &= \frac{d}{dx} \left[\ln(x + \sqrt{x^2 + 1}) \right] = \frac{1}{x + \sqrt{x^2 + 1}} \left(1 + \frac{x}{\sqrt{x^2 + 1}} \right) \\ &= \frac{1}{x + \sqrt{x^2 + 1}} \left(\frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1}} \right) = \frac{1}{\sqrt{x^2 + 1}} \end{aligned}$$

- To differentiate $y = \cosh^{-1} x$,

Let $\cosh y = x$,

By implicit differentiation,

$$\sinh y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sinh y} = \frac{1}{\sqrt{\cosh^2 y - 1}} = \frac{1}{\sqrt{x^2 - 1}}$$

- To differentiate $y = \tanh^{-1} x$,

Let $\tanh y = x$,

By implicit differentiation,

$$\operatorname{sech}^2 y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\operatorname{sech}^2 y} = \frac{1}{1 - \tanh^2 y} = \frac{1}{1 - x^2}$$

- **Example 3:**

Given that $y = \sinh^{-1} x + \tanh^{-1} x$ ($-1 < x < 1$), show that $\frac{dy}{dx} = 0$ for only one value of x .

- **Example 4:**

It is given that $y = \operatorname{cosech}^{-1} x$. Show that for $x > 0$, $\frac{dy}{dx} = -\frac{1}{x\sqrt{x^2+1}}$ and write down an expression for $\frac{dy}{dx}$ in terms of x for $x < 0$.

- **Example 5:**

Prove that $\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}$. Given that $y = (\coth^{-1} x)^2$, show that

$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} = \frac{k}{1-x^2}$, where k is a constant to be determined.

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- **Example 6:**

It is given that $y = \frac{1}{2} \ln(\coth x)$, $x > 0$. Show that $\frac{dy}{dx} = -\operatorname{cosech}(2x)$.

- **Example 7:**

Given that $y = \sinh^{-1}(\tanh x)$, show that $\frac{dy}{dx} = \frac{\operatorname{sech}^2 x}{\sqrt{1+\tanh^2 x}}$.

- **Exercise C: Differentiate the following functions.**

(i) $y = \sinh^{-1} 5x$

(ii) $y = \cosh^{-1} 3x$

(iii) $y = \sinh^{-1} \sqrt{2}x$

(iv) $y = \cosh^{-1} \frac{3}{4}x$

(v) $y = \sinh^{-1} x^2$

(vi) $y = \operatorname{sech}^{-1} x$

4. To obtain an expression for $\frac{d^2y}{dx^2}$ in the case where the relation between y and x is defined implicitly.

- **Example 8:**

Find the value of $\frac{d^2y}{dx^2}$ at the point (1, 2) on the curve $2x^3 + 2y^3 - 9xy = 0$.

- **Example 9:**

Show that $\frac{dy}{dx} = 1$ at the point $A(1,1)$ on the curve $y + x \ln y - x^2 = 0$. Find the value of $\frac{d^2y}{dx^2}$ at A .

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- **Example 10:**

Show that $\frac{dy}{dx} = -\frac{1}{2}$ at the point $A\left(1, \frac{1}{2}\right)$ on the curve $x + 2y + \tan^{-1}(2xy) = 2 + \frac{\pi}{4}$. Find the value of $\frac{d^2y}{dx^2}$ at A .

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- **Example 11:**

Given that the equation of a curve is $y = x + \tanh^{-1}(x + y)$, show that, at all points of the curve except the origin,

(i) $\frac{dy}{dx} = 1 - \frac{2}{(x+y)^2}$

(ii) $\frac{d^2y}{dx^2} = 8 \cosh^3(x - y) \operatorname{cosech}^5(x - y).$

- **Example 12 (PYQ O/N 2016/11/12/13/8):**

A curve C has equation $x^2 + 4xy - y^2 + 20 = 0$. Show that, at stationary points on C , $x = -2y$. [3]

Find the coordinates of the stationary points on C , and determine their nature by considering the value of $\frac{d^2y}{dx^2}$ at the stationary points. [8]

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- **Example 13 (PYQ O/N 2006/1/10):**

The curve C has equation

$$y = x^2 + \lambda \sin(x + y),$$

where λ is a constant, and passes through the point $A\left(\frac{1}{4}\pi, \frac{1}{4}\pi\right)$. Show that C has no tangent which is parallel to the y -axis. [5]

Show that, at A ,

$$\frac{d^2y}{dx^2} = 2 - \frac{1}{64}\pi(4 - \pi)(\pi + 2)^2. \quad [5]$$

- **Exercise D:**

1. The curve C has equation $xy - \ln(x + y^2 - 10) - 6 = 0$.
 - (i) Show that, at any point of C , $\left(x \frac{dy}{dx} + y\right)(x + y^2 - 10) = 1 + 2y \frac{dy}{dx}$.
 - (ii) Find the value of $\frac{d^2y}{dx^2}$ at the point $(2, 3)$.
2. A curve has equation $(x^2 + y^2)^2 = 16(x^2 - y^2)$.
 - (i) Show that $(y^3 + 8y + x^2y) \frac{dy}{dx} = 8x - xy^2 - x^3$.
 - (ii) Find the coordinates of the stationary points of the curve.
 - (iii) Find the coordinates of the points where the tangent is vertical.
 - (iv) By finding the second derivative, classify the stationary points.

3. A curve has equation $\sin 2x + \sin y = 1$.

(i) Find the gradient at the point where $x = \frac{\pi}{3}$.

(ii) Find the value of $\frac{d^2y}{dx^2}$ at this point.

4. A curve has equation $xe^{\sin y} = 3$.

(i) Find an expression for $\frac{d^2y}{dx^2}$.

(ii) Find the value of $\frac{d^2y}{dx^2}$ when $x = 4$.

5. To obtain an expression for $\frac{d^2y}{dx^2}$ in the case where the relation between y and x is defined parametrically.

- If a curve is defined by the parametric equations $x = f(t), y = g(t)$, then

$$\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right), \text{ where } \frac{dy}{dx} \text{ is a function of } t$$

$$= \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx}$$

- **Example 14:**

A curve is defined parametrically by $x = t^2 - 2 \ln t, y = 4(t - 1), t \geq 1$. Find $\frac{d^2y}{dx^2}$ in terms of t .

- **Example 15:**

A curve is defined parametrically by $x = \cos 2t$, $y = 2t + \sin 2t$. Show that $\frac{dy}{dx} = -\cot t$ and hence determine the value of $\frac{d^2y}{dx^2}$ in terms of t .

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- **Example 16:**

Given that $x = a(t - \sin t)$ and $y = a(1 - \cos t)$ where a is a positive constant, find $\frac{dy}{dx}$ in terms of t , and show that $\frac{d^2y}{dx^2} = -\frac{1}{4a} \operatorname{cosec}^4\left(\frac{1}{2}t\right)$.

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- **Example 17:**

The curve C is defined parametrically for all real values of t by the equations
 $x = t + \cosh t, y = \sinh t$.

(a) Show that $\frac{d^2y}{dx^2} = \frac{\sinh(t)-1}{(\sinh(t)+1)^3}$.

(b) Hence find the coordinates of the point on C at which $\frac{d^2y}{dx^2} = 0$.

- **Example 18:**

The curve C has parametric equations $x = \sin t, y = \sin 2t$ for $0 \leq t \leq \pi$.

Find $\frac{d^2y}{dx^2}$ in terms of t .

Hence, or otherwise, find the coordinates of the stationary points on C and determine their nature.

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- **Example 19 (PYQ O/N 2009/1/4):**

It is given that

$$x = t + \sin t, \quad y = t^2 + 2 \cos t,$$

where $-\pi < t < \pi$. Find $\frac{dy}{dx}$ in terms of t . [2]

Show that

$$\frac{d^2y}{dx^2} = \frac{2t \sin t}{(1 + \cos t)^3}. \quad [4]$$

Show that $\frac{dy}{dx}$ increases with x over the given interval of t . [2]

- **Example 20 (PYQ O/N 2013/13/4):**

A curve has parametric equations

$$x = 2\theta - \sin 2\theta, \quad y = 1 - \cos 2\theta, \quad \text{for } -3\pi \leq \theta \leq 3\pi.$$

Show that

$$\frac{dy}{dx} = \cot \theta,$$

except for certain values of θ , which should be stated. [4]

Find the value of $\frac{d^2y}{dx^2}$ when $\theta = \frac{1}{4}\pi$. [3]

- **Exercise E:**

1. A curve is defined by the parametric equations $x = t^2 - t + 1, y = t^3 - 3t - 1$.
 - (a) Find the coordinates of the stationary points of the curve.
 - (b) Identify the nature of these stationary points.
 - (c) Show that the curve passes through one of stationary points twice.
2. A curve is defined by the parametric equations $x = t + \sin 2t, y = \cos 2t$.
 - (a) Show that $\frac{dy}{dx} = \frac{-2 \sin 2t}{1 + \cos 2t}$.
 - (b) Show that $\frac{d^2y}{dx^2} = \frac{-4(y+2)}{(1+2y)^3}$.

3. A curve is defined by the parametric equations $x = 1 + 2 \cos t$, $y = \cos t + \sin t$.
- (a) Show that $\frac{dy}{dx} = \frac{\tan(t)-1}{2 \tan t}$.
- (b) Show that $\frac{d^2y}{dx^2} = \frac{-\sec^2 t}{4 \tan^3 t}$.
- (c) Hence find and classify all the stationary points of the curve.
4. The curve C is defined parametrically by $x = \frac{3t}{1+t^2}$, $y = \frac{3t^2}{1+t^3}$, where t is a real parameter and $t \neq -1$. Given that $t \neq 2^{-\frac{1}{3}}$, show that $\frac{dy}{dx} = \frac{2t-t^4}{1-2t^3}$.
- Show also that there is no point of C at which $\frac{d^2y}{dx^2} = 0$.



6. Maclaurin's Series

- If $f(x)$ can be expanded in a convergent series of positive integral powers of x , then

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots$$

- **Proof:** Let $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots$

Differentiating repeatedly with respect to x gives

$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \dots$$

$$f''(x) = 2a_2 + 6a_3x + 12a_4x^2 + \dots$$

$$f'''(x) = 6a_3 + 24a_4x + \dots$$

$$a_0 = f(0); a_1 = f'(0); a_2 = \frac{1}{2}f''(0) = \frac{1}{2!}f''(0); a_3 = \frac{1}{6}f'''(0) = \frac{1}{3!}f'''(0); \dots$$

$$\therefore f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots$$

- **Example 21:** Find Maclaurin series for e^x .



- **Example 22:** Find Maclaurin series for $\ln(1 + x)$.

- **Example 23:** Find Maclaurin series for $\sin x$.



- **Example 24:** Find Maclaurin series for $\cos x$.

Standard Expansion (in MF19)

Maclaurin's series:

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^r}{r!}f^{(r)}(0) + \dots$$

$$e^x = \exp(x) = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^r}{r!} + \dots \quad (\text{all } x)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + (-1)^{r+1} \frac{x^r}{r} + \dots \quad (-1 < x \leq 1)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^r \frac{x^{2r+1}}{(2r+1)!} + \dots \quad (\text{all } x)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^r \frac{x^{2r}}{(2r)!} + \dots \quad (\text{all } x)$$

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots + (-1)^r \frac{x^{2r+1}}{2r+1} + \dots \quad (-1 \leq x \leq 1)$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + \frac{x^{2r+1}}{(2r+1)!} + \dots \quad (\text{all } x)$$

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots + \frac{x^{2r}}{(2r)!} + \dots \quad (\text{all } x)$$

$$\tanh^{-1} x = x + \frac{x^3}{3} + \frac{x^5}{5} + \dots + \frac{x^{2r+1}}{2r+1} + \dots \quad (-1 < x < 1)$$

- If $f(x)$ is undefined at $x = 0$, i.e. $f(0) = \infty$ or $-\infty$, then $f(x)$ cannot be expressed as a Maclaurin series.
- Notice that $\frac{1}{x}$, $\ln x$ and $\cot x$ cannot be expressed by Maclaurin series because they are undefined at $x = 0$.
- There are two ways to find Maclaurin series of certain function.
 - (i) Repeat the process step by step as in the previous examples.
 - (ii) With the help of MF19.
- **Example 25:**

If $y = \tan(e^x - 1)$, prove that $\frac{d^2y}{dx^2} = \frac{dy}{dx}(1 + 2e^xy)$. Obtain the Maclaurin expansion of y as far as the term in x^3 .

- **Example 26:**

If $y = \ln(\sec x + \tan x)$, prove that $e^y \frac{dy}{dx} = \left(\frac{dy}{dx}\right)^2 + \frac{d^2y}{dx^2}$. Expand y in ascending powers of x as far as the term in x^3 .

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- **Example 27:**

It is given that $f(x) = \tanh^{-1}\left(\frac{1-x}{3+x}\right)$ for $x > -1$.

(a) Show that $f''(x) = \frac{1}{2(x+1)^2}$.

(b) Hence find the Maclaurin series for $f(x)$ up to and including the term in x^2 .

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- **Example 28:**

If $y = (\sin^{-1} x)^2$ and $-\frac{1}{2}\pi < x < \frac{1}{2}\pi$, Show that $(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 2$. Obtain the expansion of y as far as the term in x^4 .

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- **Example 29:**

It is given that $y = \sinh(2 \sinh^{-1} x)$, show that

(a) $(1 + x^2) \left(\frac{dy}{dx} \right)^2 - 4y^2 = 4$

(b) $(1 + x^2) \frac{d^2y}{dx^2} + x \frac{dy}{dx} - 4y = 0.$

Hence expand y in ascending powers of x as far as the term in x^5 .

- **Exercise F:**

1. Given that $y = e^{\sin^{-1} x}$, prove that $(1 - x^2) \frac{d^2 y}{dx^2} = y + x \frac{dy}{dx}$. Hence expand y in ascending powers of x up to the term in x^3 .
2. Given that $y = \cos^{-1} x$, prove that $(x^2 - 1) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = 0$. Hence find the first three non-zero terms in series expansion for $\cos^{-1} x$.

3. Given that $y = e^{\sqrt{1+x}}$, prove that $4(1+x)\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = y$. Hence expand y in ascending powers of x up to the term in x^4 .
4. Given that $y = e^{-3x} \sin 2x$, show that $\frac{dy}{dx} = 2e^{-3x} \cos 2x - 3y$ and $\frac{d^2y}{dx^2} = -13y - 6\frac{dy}{dx}$. Hence, find the Maclaurin series for y up to the term in x^4 .

5. Given that $y = \tan\left(2x + \frac{4}{3}\pi\right)$, show that $\frac{dy}{dx} = 2(1 + y^2)$ and $\frac{d^3y}{dx^3} = 4\left[\left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2}\right]$. Hence, find the Maclaurin series for y up to the term in x^3 .
6. Given that $y = \frac{1}{1+\sin 2x}$, find the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when $x = 0$. Hence find the first three non-zero terms in series expansion for y .



7. Find the Maclaurin series expansion for y up to and including the term x^2 given that $\frac{dy}{dx} = y^3 + x^8$ and $y = 1$ when $x = 0$.

- 8.(a) Find the first three non-zero terms of the Maclaurin series for

$$\frac{e^{-2x} \sin x}{\ln(1+x)}.$$

Then, state the range of values of x for which the series is valid.

- (b) Using the result that $\tan^{-1} x = \int_0^x \frac{1}{1+t^2} dt$, show that $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$, where $|x| < 1$. Hence

$$\lim_{x \rightarrow 0} \frac{\tan^{-1} x}{\sin 2x}.$$